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Pose landmark detection guide

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The MediaPipe Pose Landmarker task lets you detect landmarks of human bodies in an image or video. You can use this task to identify key body locations, analyze posture, and categorize movements. This task uses machine learning (ML) models that work with single images or video. The task outputs body pose landmarks in image coordinates and in 3-dimensional world coordinates.



Try it! →

Get Started

Start using this task by following the implementation guide for your target platform. These platform-specific guides walk you through a basic implementation of this task, including a recommended model, and code example with recommended configuration options:

- Android** - [Code example](#) - [Guide](#)
- Python** - [Code example](#) - [Guide](#)
- Web** - [Code example](#) - [Guide](#)

Task details

This section describes the capabilities, inputs, outputs, and configuration options of this task.

Features

- Input image processing** - Processing includes image rotation, resizing, normalization, and color space conversion.
- Score threshold** - Filter results based on prediction scores.

Task inputs	Task outputs
The Pose Landmarker accepts an input of one of the following data types: - Still images - Decoded video frames - Live video feed	The Pose Landmarker outputs the following results: - Pose landmarks in normalized image coordinates - Pose landmarks in world coordinates - Optional: a segmentation mask for the pose.

Configurations options

This task has the following configuration options:

Option Name	Description	Value Range	Default Value
running_mode	Sets the running mode for the task. There are three modes: IMAGE: The mode for single image inputs. VIDEO: The mode for decoded frames of a video. LIVE_STREAM: The mode for a livestream of input data, such as from a camera. In this mode, resultListener must be called to set up a listener to receive results asynchronously.	{IMAGE, VIDEO, LIVE_STREAM}	IMAGE
num_poses	The maximum number of poses that can be detected by the Pose Landmarker.	Integer > 0	1
min_pose_detection_confidence	The minimum confidence score for the pose detection to be considered successful.	Float [0.0, 1.0]	0.5
min_pose_presence_confidence	The minimum confidence score of pose presence score in the pose landmark detection.	Float [0.0, 1.0]	0.5
min_tracking_confidence	The minimum confidence score for the pose tracking to be considered successful.	Float [0.0, 1.0]	0.5
output_segmentation_masks	Whether Pose Landmarker outputs a segmentation mask for the detected pose.	Boolean	False
result_callback	Sets the result listener to receive the landmarker results asynchronously when Pose Landmarker is in the live stream mode. Can only be used when running mode is set to LIVE_STREAM	Result Listener	N/A

Models

The Pose Landmarker uses a series of models to predict pose landmarks. The first model detects the presence of human bodies within an image frame, and the second model locates landmarks on the bodies.

The following models are packaged together into a downloadable model bundle:

- Pose detection model**: detects the presence of bodies with a few key pose landmarks.
- Pose landmarker model**: adds a complete mapping of the pose. The model outputs an estimate of 33 3-dimensional pose landmarks.

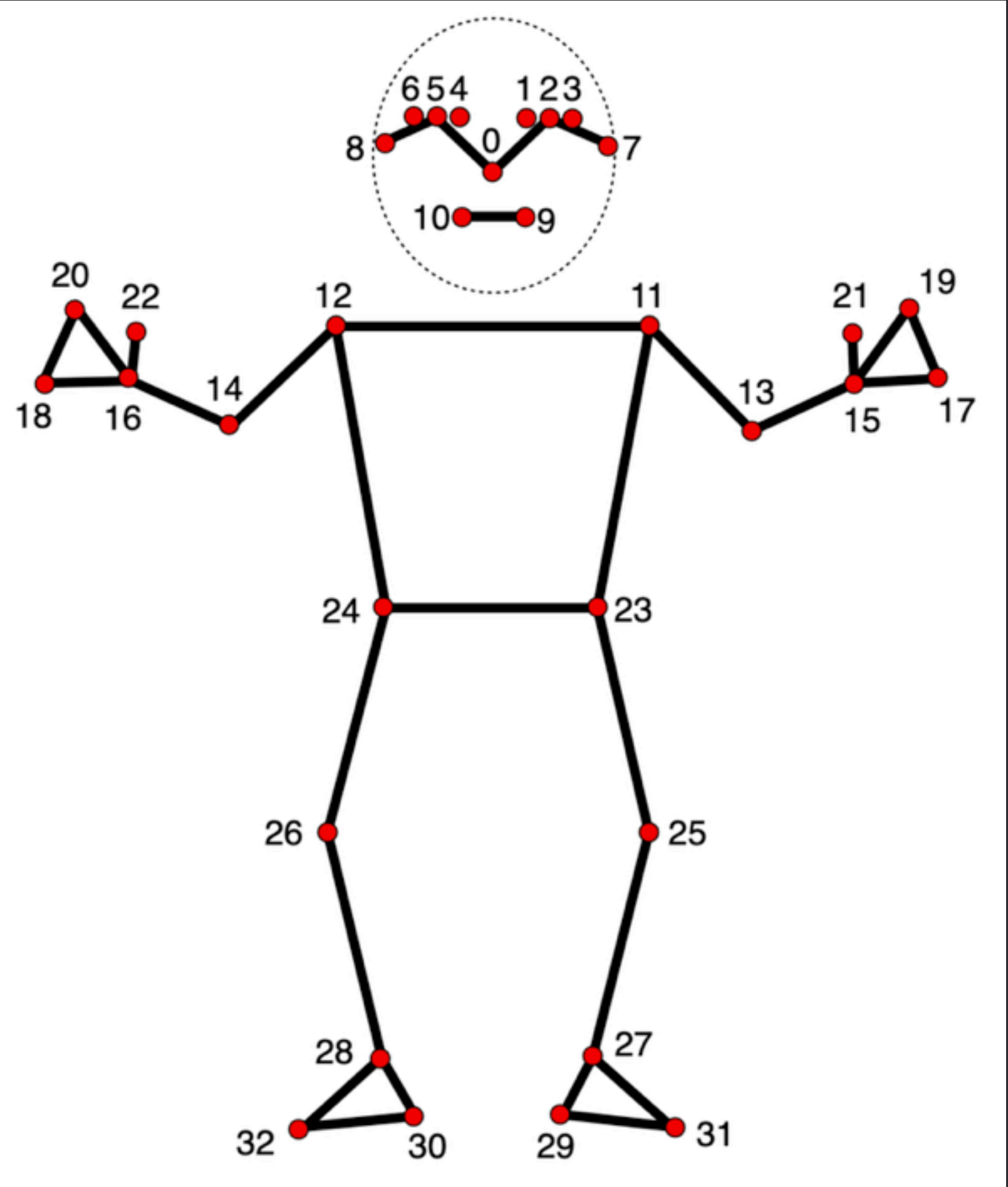
This bundle uses a convolutional neural network similar to [MobileNetV2](#) and is optimized for on-device, real-time fitness applications. This variant of the [BlazePose](#) model uses [GHUM](#), a 3D human shape modeling pipeline, to estimate the full 3D body pose of an individual in images or videos.

Attention: This MediaPipe Solutions Preview is an early release. [Learn more](#).

Model bundle	Input shape	Data type	Model Cards	Versions
Pose landmarker (lite)	Pose detector: 224 x 224 x 3 Pose landmarker: 256 x 256 x 3	float 16	info	Latest
Pose landmarker (Full)	Pose detector: 224 x 224 x 3 Pose landmarker: 256 x 256 x 3	float 16	info	Latest
Pose landmarker (Heavy)	Pose detector: 224 x 224 x 3 Pose landmarker: 256 x 256 x 3	float 16	info	Latest

Pose landmarker model

The pose landmarker model tracks 33 body landmark locations, representing the approximate location of the following body parts:



- 0 - nose
- 1 - left eye (inner)
- 2 - left eye
- 3 - left eye (outer)
- 4 - right eye (inner)
- 5 - right eye
- 6 - right eye (outer)
- 7 - left ear
- 8 - right ear
- 9 - mouth (left)
- 10 - mouth (right)
- 11 - left shoulder
- 12 - right shoulder
- 13 - left elbow
- 14 - right elbow
- 15 - left wrist
- 16 - right wrist
- 17 - left pinky
- 18 - right pinky
- 19 - left index
- 20 - right index
- 21 - left thumb
- 22 - right thumb
- 23 - left hip
- 24 - right hip
- 25 - left knee
- 26 - right knee
- 27 - left ankle
- 28 - right ankle
- 29 - left heel
- 30 - right heel
- 31 - left foot index
- 32 - right foot index

The model output contains both normalized coordinates (**Landmarks**) and world coordinates (**WorldLandmarks**) for each landmark.

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